Notes: Baker, Larcker & Wang (JFE, 2022)

How Much Should We Trust Staggered Difference-in-Differences Estimates?

Contents

1 Big picture

- **Question.** When should applied researchers trust staggered difference-in-differences estimates obtained from two-way fixed effects regressions?
- **Main answer.** Standard static and event-study TWFE estimators can be badly biased when treatment timing is staggered and treatment effects are heterogeneous across cohorts or over time.
- **Why this matters.** Finance, accounting, and law papers often exploit staggered adoption of laws, regulations, or market rules across states, countries, industries, or firms. The paper documents that more than half of DiD papers in top finance and accounting journals from 2000–2019 use staggered designs.
- **Core econometric problem.** With staggered treatment timing, already-treated units can serve as controls for later-treated units. If treatment effects are dynamic, the outcome changes of already-treated units include treatment effects. These are "bad comparisons."
- **Static TWFE implication.** The coefficient in a static TWFE DiD regression is a weighted average of many two-by-two DiD comparisons. Some comparisons use later-treated units as controls for earlier-treated units, and some use earlier-treated units as controls for later-treated units. The weights need not correspond to the target average treatment effect on treated units.
- **Event-study implication.** Dynamic event-study coefficients do not automatically solve the problem. Under staggered timing, contaminated comparisons can create spurious pre-trends, hide real pre-trends, or distort post-treatment dynamics.
- **Simulation contribution.** Using Compustat-like data-generating processes, the paper shows that TWFE estimates are reliable in simple cases but become biased when staggered treatment timing and treatment effect heterogeneity appear together.
- **Practical contribution.** The paper explains and compares three modern alternatives: Callaway and Sant'Anna, Sun and Abraham, and stacked regressions.
- **Reanalysis contribution.** The paper reexamines two published applications: Beck, Levine, and Levkov on banking deregulation and inequality; and Fauver et al. on board reforms and firm value.
- **Takeaway.** Researchers should not treat staggered TWFE as a default causal estimator. They should diagnose the comparison structure and report robust estimators designed for staggered treatment timing and heterogeneous effects.

2 Data, empirical settings, and objects of interest

2.1 Why staggered DiD is common in finance

- Many empirical finance questions study policy or institutional changes that occur at different times across units.
- Examples include state law changes, bank deregulation, board reforms, anti-takeover laws, governance reforms, disclosure rules, and legal enforcement changes.
- Researchers typically estimate regressions with unit fixed effects and year fixed effects, interpreting the treatment coefficient as an average causal effect.
- The paper shows that this practice is widespread: among 744 top finance/accounting DiD papers from 2000–2019, 407 use staggered DiD designs.

Interpretation: The widespread use of staggered designs makes the econometric problem practically important. The paper is not just a technical warning; it directly targets research designs common in finance and accounting.

2.2 Target estimand

Let D_{it} be an indicator for whether unit i is treated in period t, and let $Y_{it}(1)$ and $Y_{it}(0)$ be treated and untreated potential outcomes. A natural estimand is the average treatment effect on the treated:

$$\text{ATT} = E\left[Y_{it}(1) - Y_{it}(0) \mid D_{it} = 1\right]. \tag{1}$$

- In simple two-period, two-group DiD, the DiD estimator recovers this target under parallel trends.
- In staggered designs, there are many cohorts and treatment periods.
- The TWFE coefficient need not equal a transparent average of cohort-time ATTs.

Interpretation: The problem is not only that parallel trends may fail. Even if treatment timing is as-if random and untreated potential outcomes satisfy parallel trends, TWFE can be biased because of contaminated comparisons.

2.3 Static TWFE specification

The standard static staggered DiD regression is

$$Y_{it} = \alpha_i + \lambda_t + \delta^{DD} D_{it} + \varepsilon_{it}. \tag{2}$$

- α_i are unit fixed effects.
- λ_t are time fixed effects.
- D_{it} is an absorbing treatment indicator.
- δ^{DD} is usually interpreted as an average treatment effect.

Interpretation: Under staggered timing, δ^{DD} is a weighted average of many two-by-two comparisons, not a simple comparison of treated units to never-treated units.

2.4 Dynamic event-study specification

A common diagnostic is an event-study regression:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - G_i = k\} + \varepsilon_{it}, \tag{3}$$

where G_i is the treatment cohort year.

- Researchers use pre-treatment coefficients to assess pre-trends.
- They use post-treatment coefficients to study dynamic treatment effects.
- In staggered designs, event-study coefficients can also be contaminated by already-treated units.

Interpretation: A clean-looking TWFE event study does not necessarily validate the research design. Likewise, apparent pre-trends can be artifacts of treatment effect heterogeneity.

3 Econometric problem and intuition

3.1 Good and bad comparisons

In staggered treatment settings, a static TWFE coefficient combines several comparisons:

1. early-treated units versus never-treated units;

2. late-treated units versus never-treated units;

3. early-treated units versus later-treated units before the latter are treated;

4. later-treated units versus already-treated early units after the latter have been treated.

- The first three can be clean if the control units are not treated during the comparison window.
- The fourth is problematic because already-treated units are used as controls.
- If treatment effects are dynamic, changes in already-treated outcomes reflect treatment effects, not untreated counterfactual trends.

Interpretation: The problem is a comparison problem. TWFE regressions do not know which comparisons are causally meaningful.

3.2 Goodman-Bacon decomposition

Goodman-Bacon's decomposition writes the TWFE coefficient as a weighted average of two-by-two DiD estimates:

$$\widehat{\delta}^{DD} = \sum_{a,b} w_{ab} \widehat{\delta}_{ab}^{2\times 2}, \tag{4}$$

where a, b index treatment cohorts and w_{ab} are weights that sum to one.

- Some $\widehat{\delta}_{ab}^{2\times 2}$ compare treated units to clean untreated or not-yet-treated units.
- Other $\widehat{\delta}_{ab}^{2\times 2}$ compare later-treated units to already-treated units.
- The latter can have the wrong sign when treatment effects grow over time.

Interpretation: A TWFE coefficient can be statistically precise but economically misleading if the weighted average is dominated by bad comparisons.

3.3 When TWFE is safe

The simulations illustrate three safe or nearly safe cases:

- a single treatment date with constant effects;
- a single treatment date with dynamic effects;
- staggered treatment timing with homogeneous constant effects across cohorts and over time.

Interpretation: Staggered timing alone is not fatal. Dynamic or heterogeneous treatment effects alone are not fatal. The problematic combination is staggered timing plus heterogeneity that makes already-treated units unsuitable controls.

3.4 When TWFE is dangerous

TWFE is likely biased when:

- treatment effects differ by cohort;
- treatment effects evolve over time since treatment;
- the treatment is absorbing and long-lasting;
- most units eventually become treated;

- no never-treated or last-treated group remains as a clean comparison group;
- event-study windows use already-treated units as effective controls.

Interpretation: These conditions are common in finance: state laws, banking deregulation, governance reforms, and international policy changes are often staggered, persistent, and eventually widespread.

4 Simulation design and results

4.1 Compustat-based simulation design

The paper constructs synthetic datasets using firm-level Compustat-like panels.

- The outcome is return on assets (ROA).
- Treatment timing mimics staggered state-level law changes.
- Simulations vary whether treatment timing is staggered and whether treatment effects are constant, dynamic, or heterogeneous.
- The target effect is known by construction, so bias is observable.

Interpretation: The simulations are designed to mimic applied corporate finance research rather than abstract stylized panels.

4.2 Simulations with uniform timing or homogeneous effects

Read:

- Figure 1 shows that TWFE works when treatment timing is uniform or effects are homogeneous.
- Simulation 1 has a single treatment time and constant effects.
- Simulation 2 has a single treatment time and dynamic effects.
- Simulation 3 has staggered timing but constant/equal effects.
- In these cases, TWFE estimates are centered near the target effect.

Economic message: TWFE is not inherently invalid. It fails when the comparison structure interacts with heterogeneous dynamic treatment effects.

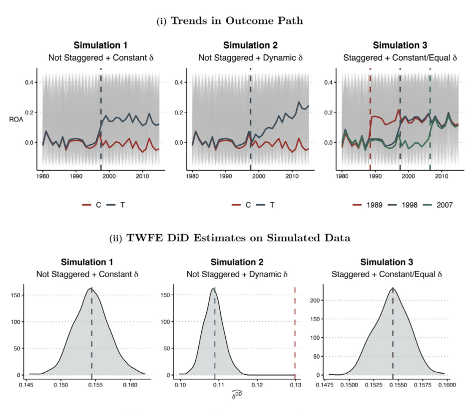


Fig. 1. Simulation: TWFE DiD Estimates Under Uniform Treatment Timing or Treatment Effect Homogeneneity.
Figure 1, panel (i), plots the firm-level outcome path (the gray lines) and the average outcome path by treatment groups (the bold lines) in one of the simulated Compustat datasets for Simulations 1, 2, and 3. To construct a simulated panel dataset, for each year, firm, and observation in the sample, we draw year-fixed effects, firm-fixed effects, and ROA residuals, respectively, from the empirical distribution. We then randomly draw states of incorporation for each firm and randomly assign states into treatment (T) and control groups (C) (i.e., in Simulations 1 and 2) or different treatment timing groups (i.e., in Simulation 3). Finally, we introduce treatment effects to the firms incorporated in treated states. Simulation 1 introduces a single treatment with a static

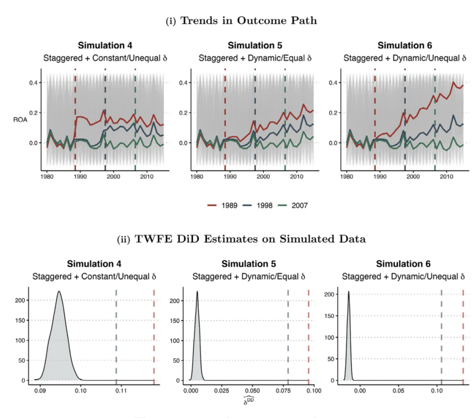


Fig. 2. Simulation: TWFE DiD Estimates Under Uniform Treatment Timing or Treatment Effect Homogeneneity.
Figure 2, panel (i), plots the firm-level outcome path (the gray lines) and the average outcome path by treatment groups (the bold lines) in one of the simulated Compustat datasets for Simulations 4, 5, and 6. To construct a simulated panel dataset, for each year, firm, and observation in the sample, we draw year-fixed effects, firm-fixed effects, and ROA residuals, respectively, from the empirical distribution. We then randomly draw states of incorporation for each firm and randomly assign states into different treatment timing groups: 1989, 1998, or 2007. Finally, we introduce treatment effects to the firms

4.3 Simulations with staggered timing and heterogeneous dynamic effects

Read:

- Figure 2 introduces staggered treatment timing with heterogeneous or dynamic treatment effects.
- TWFE estimates can differ sharply from observation-weighted or firm-weighted average treatment effects.
- In the most problematic simulation, the TWFE estimate can have the opposite sign of the true average effect.

Economic message: Bad comparisons can do more than attenuate estimates. They can generate sign reversals and false substantive conclusions.

4.4 Goodman-Bacon diagnostics

Read:

- Figure 3 decomposes the problematic TWFE estimate into constituent two-by-two comparisons.
- Earlier-versus-later and later-versus-earlier comparisons receive different weights.
- Some comparisons estimate effects with signs opposite to the true average treatment effect.
- The bottom panel illustrates how already-treated units distort the comparison of trends.

Economic message: The decomposition makes the bias visible. Researchers should inspect whether clean comparisons or already-treated comparisons dominate their estimate.

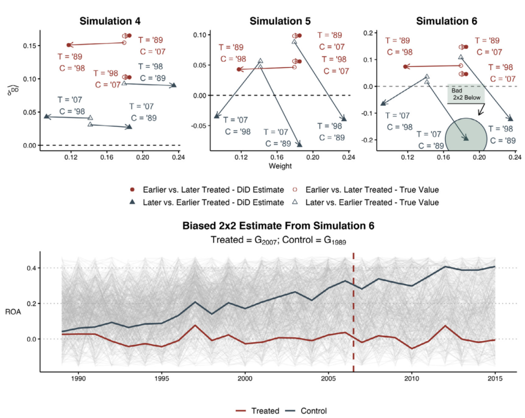
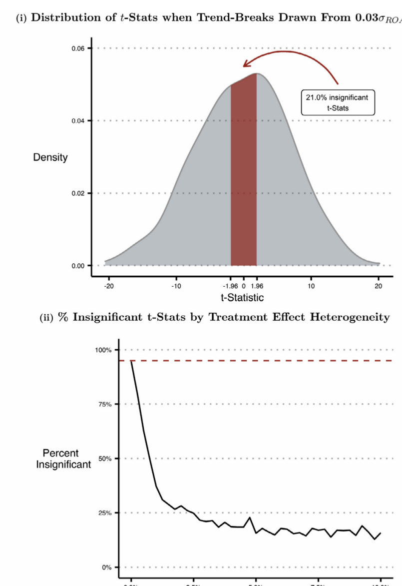


Fig. 3. Simulation: Diagnostics.
Figure 3, upper panel, plots the implicit weight given to each constituent 2×2 in the static TWFE DiD estimate and the effect estimate for that 2×2 in one of the simulated Compustat datasets for Simulation 4, 5, and 6. The solid red circles represent the empirical estimates and TWFE weights for 2×2s using later-treated firms as effective comparisons. The blue triangles represent the empirical estimates and TWFE weights for 2×2s using prior-treated firms as effective comparisons. The hollow circle or triangle represents the firm-average ATT for the treated firms in the corresponding 2×2s. The bottom panel depicts one constituent ("potentially problematic") 2×2 comparison—a comparison of firms treated in 2007 to firms treated in 1989 in Simulation 6. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

treated groups (as effective controls).[5] Of particular con- (as treatment) to the later-treated (as effective controls),



4.5 Type-I error and over-rejection

Read:

- Figure 4 studies a setting where the expected ATT is zero.
- TWFE can produce statistically significant estimates even when the true average effect is zero.
- The degree of over-rejection depends on treatment effect heterogeneity, treatment timing, and standard error behavior.

Economic message: The problem is not only biased point estimates. It also affects inference, generating false positives.

4.6 Event studies and spurious pre-trends

Read:

- Figure 5 shows that TWFE event-study estimates can display false pre-trends when there are no real pre-trends.
- Conversely, TWFE can hide real pre-trends in other settings.
- Binning, normalization, and omitted event periods can alter the shape of the event-study plot.

Economic message: Researchers should not rely only on visual inspection of TWFE event-study coefficients to validate parallel trends.

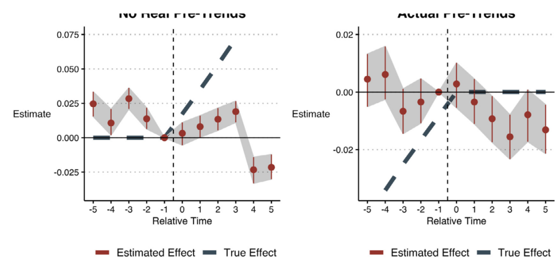


Fig. 5. Simulation: TWFE Dynamic Treatment Effect Estimates with No Real Pre-Trends and Actual Pre-Trends.
Figure 5, left-hand panel, plots the distribution of event-study estimates based on a variant of Simulation 6 (Eq. (10)) in which there are no pre-period trends and post-treatment trend-breaks of the three different cohorts are $\delta_{1989} = 0.10\sigma_{ROA}$, $\delta_{1998} = 0.05\sigma_{ROA}$, and $\delta_{1998} = 0.01\sigma_{ROA}$, where σ_{ROA} is the empirical *ROA* standard deviation in Compustat. For each of the 500 simulated Compustat panel datasets, we estimate a TWFE event-study specification (Eq. (12)) that includes relative-time indicators for the five years before and after the year of treatment (Relative Time = 0). We exclude the relative-time indicator for the year prior to treatment (Relative Time = -1). Moreover, we combine relative-time periods more than five years before treatment into one bin and relative-time periods more than five years after treatment into another bin. For each relative-time period from -5 to 5, we plot the point estimate (the solid circle), the 95% confidence interval (the vertical lines intersecting the solid circles), and the observation-average ("true") ATT for each relative-time period (the dashed line). The right-hand panel plots the distribution of event-study estimates based on Simulation 7 (Eq. (15)), which has pre-treatment trends and but no treatment effects. Aside from the data generating function, all other aspects of this simulation are the same as the left-hand-side panel.

5 Alternative estimators

5.1 Callaway and Sant'Anna estimator

The Callaway and Sant'Anna approach estimates group-time treatment effects:

$$ATT(g, t) = E[Y_t(1) - Y_t(0) \mid G = g]. \tag{5}$$

- Treatment cohorts are indexed by first treatment date g.
- For each treated cohort and time period, the method uses appropriate clean controls.
- Controls can be never-treated or not-yet-treated units, depending on assumptions and sample structure.
- Group-time effects can be aggregated into static or dynamic summaries.

Interpretation: This estimator makes the comparison group explicit and avoids using already-treated units as controls.

5.2 Sun and Abraham estimator

Sun and Abraham propose an interaction-weighted event-study estimator.

- It estimates cohort-specific relative-time effects.
- It avoids contaminating one relative-time coefficient with treatment effects from other relative-time periods.
- It is especially useful for event-study plots under heterogeneous dynamic treatment effects.

Interpretation: The Sun–Abraham estimator is designed to fix the problem that TWFE event-study coefficients can mix effects from the wrong event times.

5.3 Stacked regression approach

The stacked regression approach builds separate event-study datasets around each treated cohort.

- For each cohort, the researcher stacks treated observations with clean controls within a chosen event window.
- Clean controls are never-treated or not-yet-treated units.
- The stacked data are then analyzed with fixed effects.
- The method is flexible and often easy to implement in applied work.

Interpretation: Stacking is intuitive because it forces each event window to use only valid controls. Its main tradeoff is that results depend on window choice and stacking design.

5.4 Simulation comparison of robust estimators

Read:

- Figure 6 compares static estimates from Callaway–Sant'Anna, Sun–Abraham, and stacked regressions.
- All three are centered closer to the true effect than the problematic TWFE estimator.
- Figure 7 compares dynamic event-study estimates from the same estimators.
- The robust estimators recover the treatment path more accurately than TWFE.

Economic message: There is no single universally best estimator, but all three alternatives directly address the bad-comparison problem that undermines TWFE.

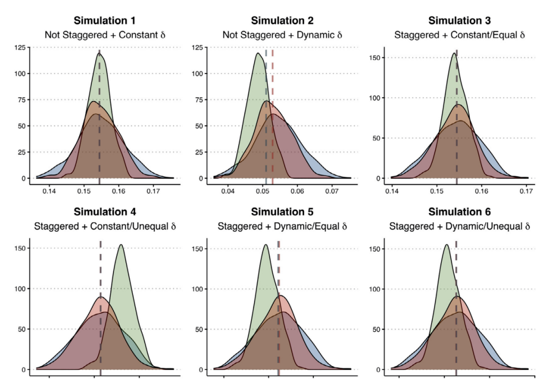


Figure 6: Static estimates from robust estimators

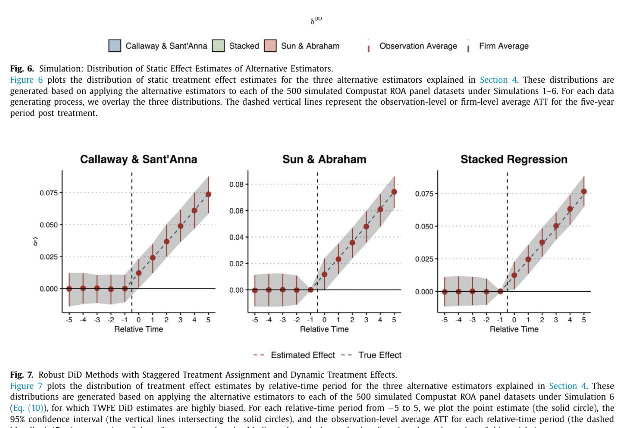


Figure 7: Dynamic estimates from robust estimators

6 Application 1: Beck, Levine, and Levkov on bank deregulation

6.1 Original research setting

- Beck, Levine, and Levkov study whether interstate bank branching deregulation reduced income inequality.
- Treatment is the timing of state-level deregulation.
- The outcome is the log of the Gini index.

- Deregulation is staggered across states, and by the late sample most states are treated.

Interpretation: This is a classic staggered policy setting in which bad comparisons can matter because early-treated states may serve as controls for later-treated states.

6.2 Replicated TWFE and alternative estimates

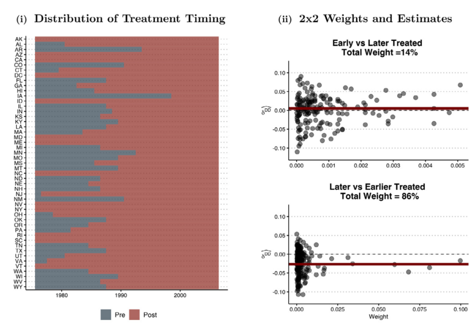
Read:

- Table 2, Panel A replicates the original TWFE estimates.
- The TWFE coefficient is negative and statistically significant, consistent with deregulation reducing inequality.
- Panel B reports robust alternatives.
- Callaway–Sant'Anna and stacked regressions produce estimates near zero and statistically insignificant.

Economic message: The substantive conclusion changes. The original TWFE result suggests a negative inequality effect, but robust estimators do not.

Panel A: Replication using TWFE		
	No Controls	With Controls
	(1) Log Gini	(2) Log Gini
Bank deregulation	-0.022^{***} (0.008)	-0.018^{***} (0.006)
Observations	1519	1519
Adj. R2	0.51	0.54
Panel B: Alternative Estimators		
	Callaway & Sant'Anna	Stacked Regressions
	(1) Log Gini	(2) Log Gini
Bank deregulation	0.001 (0.007)	0.000 (0.005)

Note: Table 2, panel A, replicates the TWFE estimates of the effects of banking deregulation on inequality (using the natural logarithm of the Gini index as a proxy) from Table II of Beck et al. (2010). The regression includes state- and year-fixed effects, and standard errors are clustered at the state level. We report the results with and without controls found in Table II of their paper. Panel B reports static effect estimates from Callaway and Sant'Anna (2021) and the stacked regression approach, using treatment observations from five years before to ten years after the year of treatment, consistent with the event-study estimates in Fig. 10, and their clean controls (not-yet-treated observations). *,**, and *** denote two-tailed significance tests at the 10%, 5%, and 1% levels, respec-



(iii) **Overall Weights and Estimates, by Timing Type**

Type	Weighted Average	Total Weight
Earlier vs Later Treated	0.005	0.143
Later vs Earlier Treated	-0.027	0.857

Fig. 8. BLL: Treatment-Timing Plots and Goodman-Bacon Decomposition Diagnostic.
Figure 8, panel (i), plots the timing of banking deregulation across states in the Beck et al. (2010) sample. Blue tiles represent pre-deregulation observations, and red tiles represent post-deregulation observations. Panel (ii) plots the TWFE weights and 2×2 DiD estimates for each treatment-timing cohort, broken down by early- (as treatment) vs. later-treated states (as controls) comparisons (in the upper half of the panel) and later- (as treatment) vs. earlier-treated states (as controls) comparisons (in the lower half of the panel). Each dot is a unique comparison between treatment-timing cohorts (e.g., states treated in 1990 compared to states treated in 1985). Panel (iii) reports the weighted average for each comparison type (bold horizontal lines in panel (ii) plots) and the total weight applied by TWFE. The overall TWFE ATT estimate is the weighted sum of each weighted average. (For interpretation of the references to

6.3 Goodman-Bacon diagnostic for the banking application

Read:

- Figure 8 plots treatment timing and the Goodman-Bacon decomposition.
- Later-treated versus earlier-treated comparisons receive large weight.
- These comparisons produce negative average effects, while clean earlier-versus-later comparisons are much closer to zero.

Economic message: The banking deregulation estimate is driven by potentially problematic comparisons that use already-treated states as controls.

6.4 Event-study evidence for the banking application

Read:

- Figure 9 replicates and modifies the BLL TWFE event study.
- The event-study plot can appear to show post-treatment declines in inequality.
- But different normalization and binning choices change the visual pattern.
- TWFE event-study evidence does not provide a reliable picture under heterogeneous dynamics.

Economic message: The event-study plot does not rescue the TWFE interpretation; it is subject to the same contamination problem.

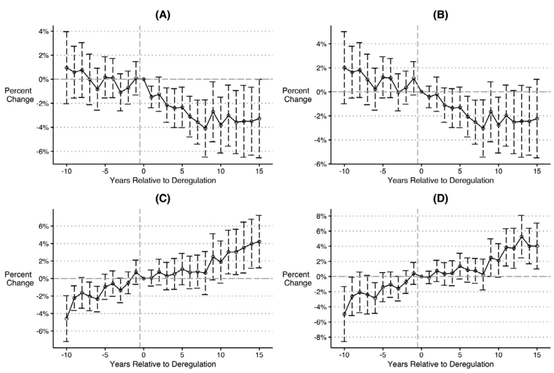


Fig. 9. BLL: TWFE DiD Event-Study Plots.
Figure 9 plots TWFE event-study estimates and 95% confidence intervals for relative-time periods from $l = g - 10$ to $l = g + 15$ around deregulation ($l = g + 0$). Panel (A) reports estimates from our replication of the event-study analysis in Beck et al. (2010). In this specification, the average value pre-adoption coefficients is subtracted from all of the event-study estimates so that the pre-period coefficients are centered at zero. Panel (B) presents estimates from a specification similar to (A) but does not subtract the average of the pre-treatment coefficients. Panel (C) presents estimates from a specification similar to (B). However, it removes the binning of relative-time periods more than ten years prior to deregulation and the binning of relative-time periods

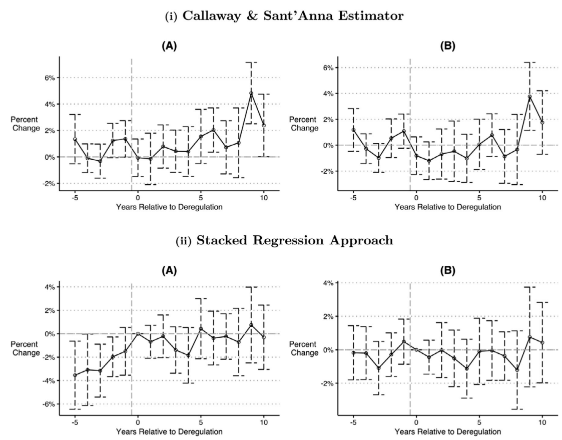


Fig. 10. BLL: CS Estimator and Stacked Regression Event-Study Plots.

6.5 Robust event studies for the banking application

Read:

- Figure 10 applies Callaway–Sant'Anna and stacked regression event-study methods.
- The robust estimates do not show strong evidence of a persistent negative effect on inequality.
- The estimates are less supportive of the original conclusion than the TWFE results.

Economic message: The banking example illustrates why robustness to modern staggered DiD estimators can be substantively important, not merely cosmetic.

7 Application 2: Fauver et al. on board reforms

7.1 Original research setting

- Fauver et al. study the effect of corporate board reforms on firm value across countries.
- Treatment is the adoption of major board reforms or first board reforms.
- The outcome is Tobin's Q.
- Treatment timing is staggered across countries.

Interpretation: This is another common finance setting: international governance reforms adopted at different times across countries.

7.2 Treatment timing and replication

Read:

- Figure 11 shows substantial variation in reform timing across countries.
- Table 3, Panel A replicates TWFE estimates.
- Major reforms and first reforms have positive and statistically significant TWFE estimates.
- Panel B reports estimates from Callaway–Sant'Anna and stacked regressions.

Economic message: The board reform application is less stark than the banking application, but robust estimators still alter inference and interpretation.

Fig. 11. FHLT: Treatment-Timing Plots.
Figure 11 plots the timing of board reforms (both the major reforms and the first reforms) across countries in the Fauver et al. (2017) sample. Blue tiles represent pre-reform observations, red tiles represent post-reform observations, and empty tiles represent missing data. The shade of the tile indicates the number of firm-year observations for each country: countries with more firm-year observations appear with darker tiles. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3

Panel A: Replication using TWFE

	With Covariates		Without Covariates	
	(1) Major Reform	(2) First Reform	(3) Major Reform	(4) First Reform
Reform	0.096*** (0.03)	0.149*** (0.05)	0.110** (0.05)	0.136** (0.07)
Control variables	Yes	Yes	No	No
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	196,016	196,016	196,016	196,016
Adj. R2	0.580	0.581	0.536	0.536

Panel B: Alternative Estimators

	Callaway & Sant'Anna		Stacked Regressions	
	(1) Major Reform	(2) First Reform	(3) Major Reform	(4) First Reform
Reform	0.062 (0.135)	0.116 (0.094)	0.063 (0.051)	0.166*** (0.055)

Note: Table 3, panel A, replicates the TWFE estimates of the effects of board reform on Tobin's Q from Table 4B of Fauver et al. (2017). The first two columns replicate the published values with firm and country covariates. In the third and fourth columns, we present the results without including covariates. All estimates use firm and year fixed effects, and robust standard errors are clustered at the country level. Panel B reports static effect estimates from Callaway and Sant'Anna (2021) and the stacked regression approach, using treatment observations from five years before to five years after the year of treatment, consistent with the event-study estimates in Fig. 13, and their clean controls (not-yet-treated observations). *,**, and *** denote two-tailed significance tests at the 10%, 5%, and 1% levels, respectively.

due to having different numbers of listed firms. In the software packages calculate clustered standard errors in

7.3 TWFE event-study plots for board reforms

Read:

- Figure 12 shows TWFE event-study estimates for major and first reforms.
- Different model specifications and event-window choices affect the pattern of coefficients.
- The plots suggest some positive post-reform effects, but they also raise concerns about contamination and specification choices.

Economic message: In staggered international reform settings, event-study specifications are sensitive to how relative-time periods are binned and normalized.

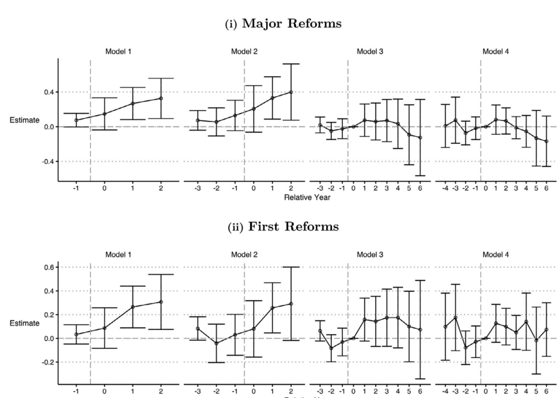


Fig. 12. FHLT: TWFE DiD Event-Study Plots.
Figure 12 plots TWFE event-study estimates and 95% confidence intervals for relative-time periods from $l = g - 4$ to $l = g + 6$ around board reform ($l = g + 1$). Panel (i) plots estimates from specifications that use major reforms as the treatment of interest. Model 1 reports estimates from our replication of the event-study analysis in Fauver et al. (2017). This specification is estimated on a truncated sample that drops observations outside of five years prior to ($l = g - 4$) or five years after ($l = g + 6$) the first year of reform adoption; it excludes all relative-time periods more than two years prior to the

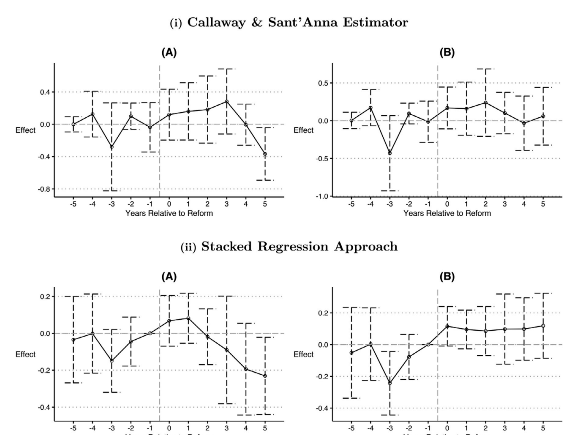


Fig. 13. FHLT: CS Estimator and Stacked Regression Event-Study Plots.

7.4 Robust event studies for board reforms

Read:

- Figure 13 reports Callaway–Sant'Anna and stacked regression event-study estimates.
- The robust estimates are smaller and often less precise than the TWFE event-study estimates.
- For first reforms, the stacked regression shows some evidence of pre-treatment movement.

Economic message: Robust estimators do not always overturn original results, but they can weaken, clarify, or complicate the original causal interpretation.

8 Practical guidance for researchers

8.1 Recommended diagnostic workflow

1. Plot treatment timing by unit or cohort.
2. Identify never-treated and last-treated units.
3. Report a Goodman-Bacon decomposition for static TWFE regressions.
4. Estimate event studies using robust alternatives, not only TWFE.
5. Report static and dynamic treatment effects using Callaway–Sant'Anna, Sun–Abraham, or stacked regressions.
6. Explain the comparison group used by each estimator.
7. Check whether results depend on event-window choices and binning.

Interpretation: The paper recommends transparency about the comparison structure rather than blind reliance on standard fixed effects regressions.

8.2 Estimator tradeoffs

- **Callaway–Sant'Anna:** flexible group-time effects; explicit control group choice; useful for aggregating cohort-time effects.
- **Sun–Abraham:** especially useful for event-study plots and heterogeneous dynamic effects.
- **Stacked regressions:** intuitive and easy to implement; forces clean controls; sensitive to stacking window and design choices.

Interpretation: The best estimator depends on the research question, available controls, treatment timing, and the estimand of interest.

8.3 What not to do

- Do not assume TWFE is valid simply because treatment timing is staggered.
- Do not interpret a single TWFE coefficient as a clean ATT without checking the decomposition.
- Do not treat TWFE event-study pre-trends as definitive evidence for or against parallel trends.
- Do not include all already-treated units as controls without considering contamination.

Interpretation: The central warning is that staggered treatment timing can make the regression use observations in ways that are not aligned with the research design.

9 Short summary

- Staggered DiD is widely used in finance and accounting.
- Standard TWFE estimators can be biased with staggered timing and heterogeneous or dynamic treatment effects.
- The static TWFE coefficient is a weighted average of many two-by-two DiD comparisons.
- Some of these comparisons use already-treated units as controls.
- These bad comparisons can create biased estimates, sign reversals, false positives, and misleading event-study plots.
- Simulations using Compustat-like data show when the problem appears and why it matters.
- Robust alternatives include Callaway–Sant'Anna, Sun–Abraham, and stacked regressions.
- Reanalysis of banking deregulation weakens the original conclusion that deregulation reduces inequality.
- Reanalysis of board reforms shows that robust estimators can materially affect magnitude and inference.
- Applied researchers should report diagnostics and robust staggered DiD estimators whenever treatment timing varies across units.

10 Conclusion

10.1 Core conclusion

Baker, Larcker, and Wang show that staggered TWFE DiD estimates often do not estimate the causal effect that researchers intend to estimate. The problem arises because already-treated units can be used as controls for later-treated units, especially when treatment effects vary across cohorts or over time. This issue is not merely theoretical; it can alter conclusions in published finance, accounting, and law applications.

10.2 Economic intuition

The intuition is simple. A clean DiD compares treated units to units that have not yet experienced treatment. Staggered TWFE regressions do not enforce this rule. When the regression compares later-treated units to already-treated units, it subtracts outcome changes that include treatment effects. If those treatment effects are dynamic, the subtraction can bias the coefficient and even flip its sign.

10.3 Takeaway

The paper's practical message is that staggered DiD requires explicit design choices. Researchers should define the target estimand, choose clean comparison groups, examine treatment timing, report decomposition diagnostics, and use robust estimators designed for heterogeneous treatment effects. TWFE can still be useful in special cases, but it should no longer be treated as the default solution for staggered treatment designs.